Project Development Phase: Snowflake Data Preparation

Data Warehouse Setup, Schema Engineering & Staging Specification

PROJECT	TARGET ENGINE	ENVIRONMENT	STATUS
Analytics Platform	**Snowflake Data Cloud**	**DEV / STAGING**	**Approved**

1. Database Architecture & Schema Initialization

The initial phase establishes virtual warehouses, dedicated role-based access control, and tiered database schemas (Raw, Staging, Analytics) to handle structured and semi-structured payloads.

Object Layer	Name Specification	Access Rights	Retention / Type
Warehouse	COMPUTE_WH_DEV	SYSADMIN, DEV_ROLE	Auto-suspend: 60s (X-Small)
Database	ENTERPRISE_DW_DEV	SYSADMIN, DATA_ENG	Standard Edition
Schema (Raw)	RAW_LANDING	INGEST_SERVICE_ROLE	Time Travel: 1 Day
Schema (Staging)	STAGING_CLEAN	TRANSFORM_ROLE	Time Travel: 7 Days
Schema (Analytics)	ANALYTICS_MART	REPORTING_ROLE, BI_USER	Time Travel: 90 Days

2. Infrastructure Provisioning Script

```
      -- 1. Create Virtual Warehouse and Database Infrastructure CREATE WAREHOUSE IF NOT EXI
STS COMPUTE_WH_DEV WITH WAREHOUSE_SIZE = 'XSMALL' AUTO_SUSPEND = 60 AUTO_RESUME = TRUE INITIA
LLY_SUSPENDED = TRUE; CREATE DATABASE IF NOT EXISTS ENTERPRISE_DW_DEV; -- 2. Create Schema Is
olation Layers USE DATABASE ENTERPRISE_DW_DEV; CREATE SCHEMA IF NOT EXISTS RAW_LANDING; CREAT
E SCHEMA IF NOT EXISTS STAGING_CLEAN; CREATE SCHEMA IF NOT EXISTS ANALYTICS_MART;
```

Key Architecture Rule: Direct mutation of RAW_LANDING tables is strictly prohibited. All raw ingested payloads are retained in immutable format to ensure complete data lineage re-playability.

3. External Staging & File Format Definition

Configuration of cloud storage integration (AWS S3 / Azure Blob) and predefined file formats for seamless batch and stream ingestion via Snowpipe.

Stage / Format	Target Source	Configuration Details	Validation Status
CSV_FORMAT	Transactional Logs	TYPE = CSV, SKIP_HEADER = 1, FIELD_OPTIONALLY_ENCLOSED_BY = ""	Validated
JSON_FORMAT	API Event Feeds	TYPE = JSON, STRIP_OUTER_ARRAY = TRUE	Validated
PARQUET_FORMAT	Clickstream Archives	TYPE = PARQUET, COMPRESSION = SNAPPY	Pending Test
S3_STAGE_RAW	s3://enterprise-data-landing/	STORAGE_INTEGRATION = S3_SNOWFLAKE_INT	Active

4. Data Cleansing & Transformation DDL

```
      -- Staging Clean Layer DDL with Data Quality Controls CREATE OR REPLACE TABLE STAGING_
CLEAN.STG_CUSTOMERS ( CUSTOMER_ID NUMBER(38,0) NOT NULL, FIRST_NAME VARCHAR(100) NOT NULL, LA
ST_NAME VARCHAR(100) NOT NULL, EMAIL VARCHAR(255) CONSTRAINT CHK_EMAIL CHECK (EMAIL LIKE '%
@%.%'), SIGNUP_TIMESTAMP TIMESTAMP_NTZ DEFAULT CURRENT_TIMESTAMP(), CUSTOMER_STATUS VARCHAR(2
0) DEFAULT 'ACTIVE', RECORD_HASH VARCHAR(64) NOT NULL, INGESTED_AT TIMESTAMP_NTZ DEFAULT CURR
ENT_TIMESTAMP() ); -- Transformation Procedure for Variant Flattening CREATE OR REPLACE VIEW
STAGING_CLEAN.VW_PARSED_EVENTS AS SELECT SRC:event_id::VARCHAR AS EVENT_ID, SRC:user_id::NUMB
ER AS USER_ID, SRC:event_type::VARCHAR AS EVENT_TYPE, TO_TIMESTAMP(SRC:timestamp::INT) AS EVE
NT_TIME FROM RAW_LANDING.RAW_EVENTS_JSON;
```

5. Quality Assurance & Audit Checks

Check Type	Target Metric / Rule	Remediation Action
Null Check	Primary keys CUSTOMER_ID, EVENT_ID cannot be NULL	Quarantine row to ERR_TABLE
Uniqueness Check	Zero duplicate records on business primary key combination	Deduplicate using ROW_NUMBER()
Referential Integrity	Foreign keys match corresponding dimension dimension table keys	Flag missing key as -1 (Unknown)